Project Title

Tree Diameter Computation and Visualization Using Dynamic Programming

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Tree Diameter Computation and Visualization System is an application designed to compute the diameter of a tree using dynamic programming techniques. The objective is to determine the longest path between any two nodes in a given tree efficiently. The system takes tree edges as input, constructs the corresponding tree structure, and processes it using a tree-based dynamic programming approach. By calculating and reusing the longest paths associated with each node, the system reduces redundant computations and efficiently determines the tree diameter.

The system produces the diameter length, representing the number of edges in the longest path between two nodes. Along with the numerical result, it visualizes the tree and highlights the diameter path, making the result easier to understand. The application is designed to handle trees with up to 1000 nodes, allowing it to process moderately large tree structures efficiently. Overall, the project demonstrates the practical use of dynamic programming, tree traversal, and visualization to solve the tree diameter problem in an efficient and intuitive manner.

Signature of the Guide